

Solving Dynamics Problems (3rd reference): Newton's Second Law

"Once is happenstance. Twice is coincidence. Three times begins to seem like a message to be ignored at one's peril".

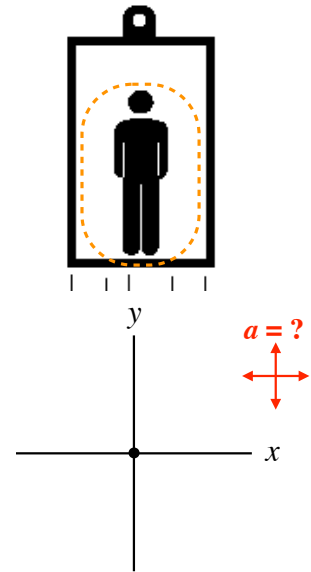
– The Orphan's of Raspay

Setup State: Sketch a Free-Body Diagram

1. Isolate the object being analyzed.
Identify ALL forces that act ON the object; NOT the forces it exerts on its surroundings.
2. Draw a *convenient* coordinate system.
3. Represent the object as a dot at the origin (particle model).
4. Draw vectors representing each of the identified forces.
(Tail of force vector on the object. Illustrate angles.)

[new]

5. Illustrate the direction of the acceleration near the FBD
(not on the center dot).



Analysis Stage: Apply Newton's Laws (i.e. "Read" the free-body diagrams).

► Similar to Newton's 1st Law problems, **EACH STEP IS GRADED.**

STEP:

EXAMPLE:

- **Physics Principle:** Write the relevant Newton's Law(s) equations(s) *independently* for each direction (axis).

i.e: Newton's 2nd Law ($\Sigma \vec{F} = m\vec{a}$)

$$\sum \vec{F}_y = m\vec{a}_y$$

- **Application:** Write the explicit sum of the vectors or component vectors along that direction.

Note: there must be a variable for every force (arrow) in the FBD.

$$\vec{F}_{1y} + \vec{F}_{2y} + \dots + \vec{F}_{Ny} = m\vec{a}_y$$

- **Sign Convention:** Separate magnitude and direction. Use +/- signs to represent the direction of the force vectors **and** the acceleration vector.

Note: the variables are now magnitudes, so no arrow caps; the direction is described by the +/- signs.

$$F_{1y} - F_{2y} + \dots - F_{Ny} = +ma_y$$

- **Trigonometry:** Write the components in terms of the original magnitudes and functions of the angles (sin or cos), when applicable. If the angle is not known, you may leave it as a component.

$$F_1 \cos \theta_1 - F_2 \sin \theta_2 + \dots = +ma_y$$

- Identify and solve for the desired quantity.

Do not carry out algebraic steps until the direction is represented with +/- signs.